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Laser level target posts

Abstract

In one example, a laser level target post for leveling a server cabinet is disclosed. The laser level target post includes an elongated body section and a reflective surface at a top end of the elongated body section. The top end reflective surface is to reflect a received light beam for leveling a top surface of the server cabinet. The laser level target post also includes a magnetic base at a bottom end of the elongated body section. The magnetic base may attach the laser level target post to a corner of the top surface of the server cabinet.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
7886450	12/2010	Fiano	33/290	G01C 15/105
8036452	12/2010	Pettersson	382/154	G01C 15/00
8209874	12/2011	Tribble	33/286	G01C 15/06
8567077	12/2012	Wernig	33/286	H01Q 3/005
8621760	12/2013	Norelli	33/DIG.1	G01C 9/28
10094076	12/2017	Fujimoto	N/A	E01C 19/004
10684124	12/2019	Hanchett	N/A	G06F 3/04815
10710751	12/2019	Monleón de la Lluvia Jordá	N/A	B64F 5/60
11253768	12/2021	Amron	N/A	G01S 13/74
11781866	12/2022	Lockhart	33/286	G01C 15/105
11828596	12/2022	Rueb	N/A	B25H 7/00
12078483	12/2023	Hammerer	N/A	G01B 11/272
12098931	12/2023	Passarini	N/A	G01S 17/42
2007/0153297	12/2006	Lau	356/620	G01C 15/006
2011/0138642	12/2010	Norelli	33/451	G01C 9/28
2013/0097880	12/2012	Wernig	33/286	H01Q 3/005
2015/0022826	12/2014	Cramer	356/620	G01B 11/14
2016/0116282	12/2015	Unger	33/228	G01S 17/88
2018/0343421	12/2017	Kahle	N/A	G01C 11/06
2022/0196403	12/2021	Hammerer	N/A	G01C 15/002
2023/0258448	12/2022	Davis	33/290	G01C 15/004
2023/0314172	12/2022	Shi	73/1.79	G01C 25/00

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
114646970	12/2021	CN	G01B 11/272
4015994	12/2021	EP	G01B 11/272

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Background/Summary

BACKGROUND

(1) Laser levels find use in the fields of construction, surveying and other like areas. A typical laser level can project a beam of laser light onto a surface for use in alignment applications. The beam of light is projected along a horizontal or vertical line or plane for use as a visual chalk line for

alignment applications. Some laser levels may be positioned on a tripod on the ground. However, the light beam from the laser is projected with reference to gravity so that the light beam plane is perfectly level even when the ground is uneven.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Examples of the disclosure will be rendered by reference to specific examples which are illustrated in the appended drawings. The drawings illustrate only particular examples of the disclosure and therefore are not to be considered to be limiting of their scope. The principles here are described and explained with additional specificity and detail through the use of the accompanying drawings.

(2) FIG. 1 illustrates a server cabinet leveling system for leveling the tops of server cabinets according to an example of the present disclosure.

(3) FIG. 2 illustrates the laser level target post of FIG. 1 for leveling a server cabinet according to an example of the present disclosure.

(4) FIG. 3 shows a cross-section of the elongated body section at point K-K of FIG. 2.

DETAILED DESCRIPTION

(5) Many data centers have uneven flooring. Such uneven flooring can arise if poured concrete is used as opposed to perforated flooring. Poured concrete flooring can vary by a few inches over a row of server cabinets which can cause the server cabinets to have different heights.

(6) A laser level may be used to align the tops of all of the server cabinets. The laser level projects a laser light beam above the server cabinets for use as a reference. A first engineer climbs a ladder to visually examine the alignment of the server cabinet tops and the line of the reference light beam. A second engineer remains on the ground to adjust the server cabinet feet/legs to increase or decrease the height of the server cabinet tops to align with the reference light beam.

(7) This process can be inefficient because it requires multiple engineers to install and align the server cabinets. Moreover, parallax error may occur when the first engineer on the ladder misjudges the alignment of a server cabinet top and the reference light beam because the alignment is being viewed from an angle. Notwithstanding this potential inaccuracy, the process can also be unsafe as it requires the use of a ladder to climb high enough to view the server cabinet tops.

(8) Accordingly, examples of the present disclosure address the foregoing by providing a laser level target post for leveling a server cabinet. The laser level target post includes an elongated body section and a reflective surface at a top end of the elongated body section. The top end reflective surface is to reflect a received light beam to facilitate leveling of a top surface of the server cabinet. Here, the light reflected by the reflective surface is visible from the ground, and the server cabinets can be adjusted without climbing a ladder. The laser level target post also includes a magnetic base at a bottom end of the elongated body section. The magnetic base may attach the laser level target post to a corner of the top surface of the server cabinet.

(9) In this manner, the server cabinet installation/alignment process is efficient because multiple engineers are not necessary to install and align the server cabinets during a data center installation. The process can be completed by a single engineer because the reflected surface is visible from the ground. Moreover, parallax error may be eliminated because the engineer on the floor can walk around and view the alignment between the reflective surfaces and the reference laser light beam from multiple angles. Further yet, this process is much safer as the engineer need not climb a ladder for alignment validation.

(10) FIG. 1 illustrates a server cabinet leveling system **100** for leveling the tops of server cabinets **104**, **106** according to an example of the present disclosure.

(11) Here, among other components, server cabinet leveling system **100** includes a set **108** of laser

level target posts **108A**, **108B**, **108C**, and **108D** as well as a laser level **112** for leveling the top of server cabinet **106**. By the term “leveling,” it is meant that a top surface **110** of server cabinet **106** and the top surfaces of adjacent server cabinets (e.g. **104**) are horizontally aligned with reference to a laser light beam **105**. Further, although set **108** in this example includes four laser level target posts **108A**, **108B**, **108C**, and **108D**, set **108** may include additional or fewer laser level target posts. Moreover, the laser level target posts on server cabinet **104** are similar to the laser level target posts **108A**, **108B**, **108C**, and **108D** but are not described for brevity.

(12) As shown, laser level target posts **108A**, **108B**, **108C**, and **108D** are positioned in respective corners of server cabinet **106**. That is, laser level target post **108A** is positioned at corner A of server cabinet **106**, laser level target post **108B** is positioned at corner B of server cabinet **106**, laser level target post **108C** is positioned at corner C of server cabinet **106** and laser level target post **108D** is positioned at corner D of server cabinet **106**. By positioning laser level target posts at each corner of server cabinet **106**, the alignment of the entirety of the top surface **110** of server cabinet **106** with the laser light beam **105** can be facilitated by adjusting a leg/foot (not shown) below the respective laser level target post.

(13) A height **H1** of all laser level target posts **108A**, **108B**, **108C**, and **108D** is the same, for example, to align the laser level target posts **108A**, **108B**, **108C**, and **108D** and the reference laser light beam **105**. An exemplary target post height **H1** may be 12 inches. Moreover, in an example, all laser level target posts are aligned in parallel with the vertical axis of their corresponding server cabinets. For example, laser level target posts **108A**, **108B**, **108C**, and **108D** are aligned in parallel with the vertical axis **109** of server cabinet **106**.

(14) Referring now to FIG. 1, server cabinet leveling system **100** further includes the laser level **112**. Laser level **112** may attach to a tripod **114** that is placed on the ground of the data center. As shown in this example, laser level **112** projects the laser light beam **105** that serves as a reference line or plane for adjusting the top surface **110** of server cabinet **106** and adjacent server cabinet top surfaces.

(15) In operation, an engineer or other user begins by placing the laser level **112** on a tripod (or on a server cabinet) as desired and then activating the laser level **112** to project the laser light beam **105**. The laser light beam **105** is projected at a predetermined height above the surface **110** of server cabinet **106**. The predetermined height at which the laser light beam **105** is projected is the same as the height **H1** of the laser level target posts **108A**, **108B**, **108C**, and **108D** above the server cabinets. The laser light beam **105** then forms a reference line or plane by which the laser level target posts **108A**, **108B**, **108C**, and **108D** are aligned. As will be further discussed with reference to FIG. 2, each laser level target post **108A**, **108B**, **108C**, and **108D** includes a top end reflective surface **116A**, **116B**, **116C**, or **116D**.

(16) While remaining on the data center floor, the engineer can observe whether the reflective surfaces **116A**, **116B**, **116C**, and **116D** are reflecting the reference laser light beam **105**. Based on this observation, the engineer can adjust the corresponding legs of the server cabinet **106**, if necessary.

(17) As an example, if reflective surface **116A** is reflecting the reference laser light beam **105**, the corner A of the surface **110** is at the correct height and needs no adjustment. In contrast, if reflective surface **116B** is not reflecting the reference laser light beam **105**, then the corner B of the surface **110** is not aligned or level. The engineer can then adjust the server cabinet foot at corner B to raise or lower corner B until reflective surface **116B** can reflect the reference laser light beam **105**.

(18) In this manner, the server cabinet installation/alignment process is efficient because multiple engineers are not necessary to install and align the server cabinets during a data center installation. The process can be completed by a single engineer. Moreover, parallax error may be eliminated because the engineer is on the floor and can walk around and view the alignment between the reflective surfaces and the reference laser light beam from multiple angles. Further yet, this process is much safer because the engineer need not climb a ladder for alignment validation.

(19) FIG. 2 illustrates the laser level target post **108A** of FIG. 1 for leveling server cabinet **106** (FIG. 1) according to an example of the present disclosure. Although not illustrated, the level target posts **108B**, **108C**, and **108D** and laser level target post **108A** are similar.

(20) In FIG. 2, laser level target post **108A** includes the reflective surface **116A** (also in FIG. 1), an elongated body section **204** and a magnetic base **206**. Reflective surface **116A** is positioned at a top end of the elongated body section **204** while the magnetic base **206** is located at a bottom end of elongated body section **204**.

(21) As noted above, top end reflective surface **116A** is to reflect the reference laser light beam **105** to aid in leveling the top surface **110** of server cabinet **106**. Reflective surface **116A** can be any reflective material consistent with the present disclosure. As an example, reflective surface **116A** can be mylar. As another example, reflective surface **116A** may be polished anodized aluminum.

(22) The reflective surface (and the laser level target post **108A**) may be part of an injection molding process, 3D printing process or the like. Or the reflective material may be attached around the periphery of the top end of laser level target post **108A** upon completion of the molding process. In FIG. 2, height H2 of reflective surface **116A** can vary depending upon the particular application. In one example, height H2 is $\frac{1}{8}$ sup.th of an inch. In another example, H2 is one inch.

(23) In FIG. 2, the elongated body section **204** connects reflective surface **116A** and the magnetic base **206**. For some examples, elongated body section **204** can be tubular as shown. Although not shown, other shapes consistent with the present disclosure may be utilized. For example, elongated body section **204** may have a polygonal shape.

(24) Elongated body section **204** may also be hollow. FIG. 3 shows a cross-section **300** of elongated body section **204** at point K-K of FIG. 2. As can be seen, in an example, elongated body section **204** is hollow with an internal diameter W3. Here, W3 may be $\frac{1}{2}$ inch, for example. A thickness T of the wall of elongated body section **204** may be $\frac{1}{16}$ inch.

(25) The elongated body section **204** provides much of the height H3 of laser level target post **108A**. By the term “elongated” it is meant that height H3 of elongated body section **204** is much greater than an external diameter W1 of the elongated body section **204**. As an example, height H3 can be 10 inches while the external diameter W1 is one inch. Thus, in this example, “elongated” means an order of magnitude of 10 to 1.

(26) Although not shown, elongated body section **204** may be adjustable to change its height as needed. In this example, elongated body section **204** may comprise two sleeves where one sleeve slides into another to adjust the height of elongated body section **204**.

(27) In FIG. 2, the magnetic base **206** is comprised of magnetic material to attach laser level target post **108A** to corner A of top surface **110** of server cabinet **106**. As shown, the height H4 of magnetic base **204** may be 1 inch. The magnetic base **204** itself has a flare having a width W2 beyond the external diameter W1 of the elongated body section **204**. In one example, width W2 is 0.5 inches on either side of elongated body section **204**.

(28) Here, note that this tubular elongated body section **204**, the top end reflective surface **116A** and the bottom end magnetic base **206** are concentric. That is, elongated body section **204**, reflective surface **116A** and magnetic base **206** are concentric about a longitudinal axis L shown in FIG. 2.

(29) As used in the description herein and throughout the claims that follow, “a”, “an”, and “the” include plural references unless the context clearly dictates otherwise. Also, as used in the description herein and throughout the claims that follow, the meaning of “in” includes “in” and “on” unless the context clearly dictates otherwise.

(30) While the above is a complete description of specific examples of the disclosure, additional examples are also possible. Thus, the above description should not be taken as limiting the scope of the disclosure which is defined by the appended claims along with their full scope of equivalents.

Claims

1. A laser level target post for leveling a server cabinet, the laser level target post comprising: an elongated body section extending in a longitudinal direction, the elongated body section having an external surface that extends in the longitudinal direction; a reflective surface at a top end of the external surface of the elongated body section, wherein the top end reflective surface is operable to reflect a received light beam for leveling a top surface of the server cabinet; and a magnetic base at a bottom end of the elongated body section, wherein the magnetic base is operable to attach the laser level target post to a corner of the top surface of the server cabinet such that the longitudinal direction extends vertically, wherein the top reflective surface is operable to reflect the received light beam if the top surface of the server cabinet is level and wherein the top reflective surface is positioned to not reflect the received light beam if the top surface of the server cabinet is not level.
2. The laser level target post of claim 1 wherein the elongated body section is tubular.
3. The laser level target post of claim 2 wherein the tubular elongated body section, the top end reflective surface and the bottom end magnetic base are concentric.
4. The laser level target post of claim 1 wherein the top end reflective surface is positioned to reflect the light beam received from a laser level.
5. The laser level target post of claim 1 wherein the top end reflective surface is to position at a predetermined vertical height from the top surface of the server cabinet.
6. The laser level target post of claim 1, wherein an axis of the elongated body section and an axis of the server cabinet are parallel.
7. The laser level target post of claim 1 wherein the reflective surface is parallel to the longitudinal direction.
8. The laser level target post of claim 1 wherein the reflective surface is arranged around an entire periphery of the top end of the external surface of the elongated body section.
9. The laser level target post of claim 1 wherein the light beam defines a height plane for leveling the top surface of the server cabinet, and the top end reflective surface is operable to reflect the light beam received when the top end reflective surface is aligned with the height plane.
10. A laser level target post for leveling a server cabinet, the laser level target post comprising: an elongated body section; a reflective surface at a top end of the elongated body section, wherein the top end reflective surface is to reflect a received light beam for leveling a top surface of the server cabinet; and a magnetic base at a bottom end of the elongated body section, wherein the magnetic base is to attach the laser level target post to a corner of the top surface of the server cabinet, wherein the top reflective surface is operable to reflect the received light beam if the top surface of the server cabinet is level, and wherein the top reflective surface is positioned to not reflect a received light beam if the top surface of the server cabinet is not level.
11. A set of laser level target posts for leveling a server cabinet, the set of laser level target posts comprising: a plurality of laser level target posts to position on corners of the server cabinet, wherein each laser level target post comprises: an elongated body section extending in a longitudinal direction, the elongated body section having an external surface that extends in the axial direction; a reflective surface at a top end of the external surface of the elongated body section; and a magnetic base at a bottom end of the elongated body section, wherein the magnetic base is operable to position the plurality of laser level target posts on each of said four corners of the server cabinet such that the longitudinal direction extends vertically, and wherein the top reflective surface is positioned to not reflect a received light beam if the top surface of the server cabinet is not level.
12. The set of laser level target posts of claim 11 wherein each one of the plurality of laser level target posts has the same length.
13. The set of laser level target posts of claim 12 wherein the tubular elongated body section, the

top end reflective surface and the bottom end magnetic base are concentric.

14. The set of laser level target posts of claim 11 wherein the elongated body section is tubular.

15. The set of laser level target posts of claim 11 wherein the top end reflective surface is positioned to reflect a light beam received from a laser level.

16. The set of laser level target posts of claim 11 wherein the top end reflective surface is to position at a predetermined vertical height from the top surface of the server cabinet.

17. The set of laser level target posts of claim 11 wherein the top reflective surface is operable to reflect a received light beam if the top surface of the server cabinet is level.

18. The set of laser level target posts of claim 11 wherein the light beam defines a height plane for leveling the top surface of the server cabinet, and the top end reflective surface of each of the plurality of laser level target posts is operable to reflect the light beam received when the top end reflective surface is aligned with the height plane.
